

Predicting Amazon stock prices using machine learning: transforming data into business insights for AWS

As a data-driven problem solver with strong analytics and storytelling skills, I bridge technical insight and business strategy to help AWS customers accelerate generative AI and predictive analytics adoption.

INTRODUCTION

Between 2020 and 2025, Amazon experienced market volatility driven by rapid innovation cycles, supply-chain disruptions, and evolving investor sentiment.

BUSINESS QUESTION

Can we predict Amazon’s future stock trends accurately enough to inform strategy, risk management, and investment timing?

BUSINESS RELEVANCE

Predictive analytics helps AWS’s finance, marketing, and investor relations teams translate complex data into actionable insights aligning with the “data-to-value” mission in enterprise AI adoption.

OBJECTIVE

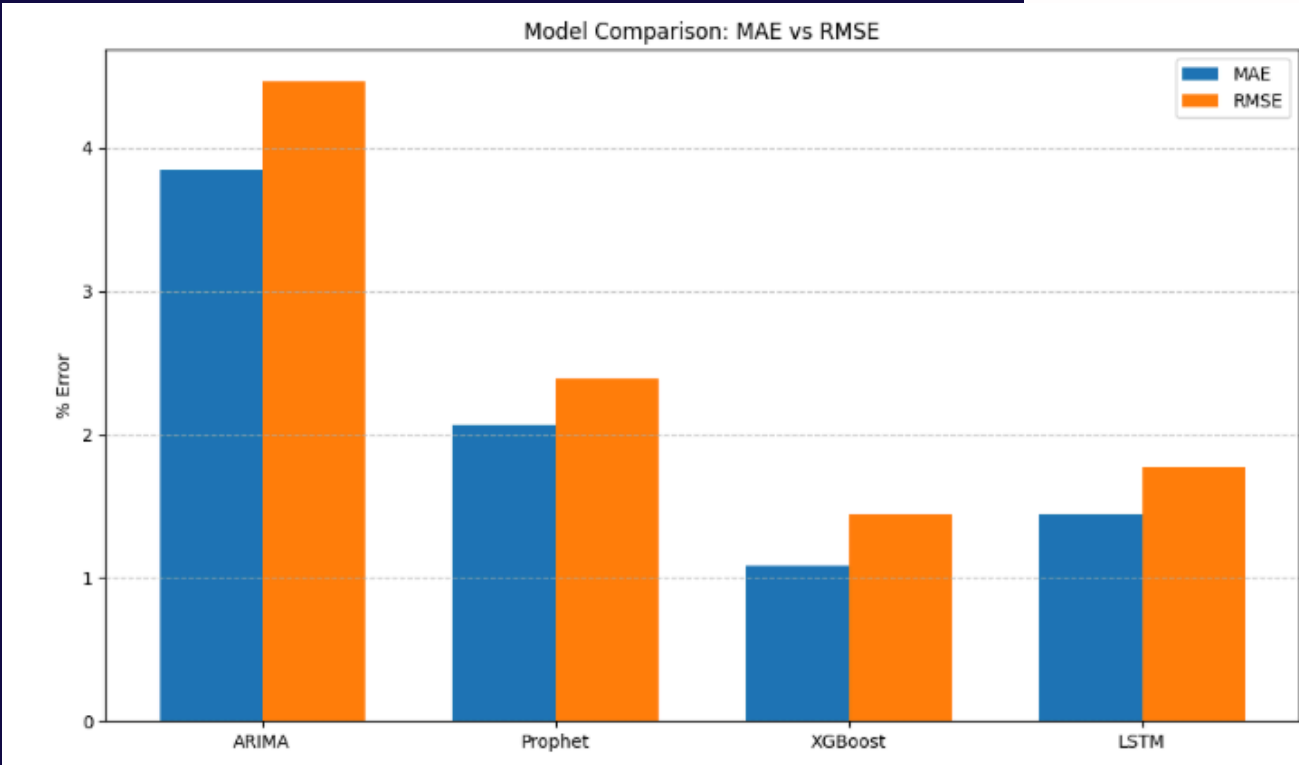
Demonstrate how machine learning can help AWS (Amazon) anticipate market dynamics and optimize strategic decisions (e.g., investment timing, sentiment analysis, revenue forecasting).

PROBLEM STATEMENT

These fluctuations challenged strategic forecasting and valuation accuracy.
This study explores how predictive modeling can improve business insight generation and guide investment or operational decisions using historical Amazon stock data.

METHODOLOGY

- Algorithms tested: Linear Regression, Random Forest, and LSTM Neural Network
- Data split 80/20 (train/test)
- Preprocessing: missing value handling, lag feature creation, moving averages, normalization



EVALUATION METRICS

Outperforms deep learning and classical models significantly.
It leverages:

- Rich technical indicators
- Engineered lag/rolling features
- Simpler training and less data-requirement compared to LSTM

ANALYSIS

LSTM, with improvement, could come closer to XGBoost
ARIMA & Prophet serves as Baselines

Great for trend + seasonality, but not for short-term fluctuations in noisy financial data
Prophet did better than ARIMA.

CONCLUSION

This demonstrates how machine learning can transform raw financial data into actionable insights for enterprise decision support.
The findings highlight AWS’s potential to accelerate predictive AI adoption through secure, scalable deployment of forecasting solutions on Bedrock and SageMaker.
By combining analytics, data strategy, and clear storytelling, the study bridges the gap between AI experimentation and business value realization.

BUSINESS VALUE TO AWS

Predictive analytics supports AWS’s goal of helping enterprises move from data to decisions.
This model exemplifies how AWS services like S3, SageMaker, and QuickSight can host scalable time-series forecasting pipelines.
Deploying similar models at enterprise scale would enable customers to:

- Anticipate market shifts and optimize investment timing
- Reduce risk through data-driven strategy
- Automate forecasting workflows in cloud-native environments